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Hyperparameter tuning techiques

Hyperparameters are learning parameters which helps in enhancing the model performance

Hyperparameter tuning its a process of selecting best parameters among all possible values

→ GridSearch CV

→ Randomised Search CV

Grid Search CV

It will search for best parameters among all possible combinations

n_estimator = [10, 20, 50, 100]

max_depth = [5, 10, 15, 20, 30]

Combinations

10, 5	20, 5	50, 5	100, 5
10, 10	20, 10	50, 10	100, 10
10, 15	20, 15	50, 15	100, 15
10, 20	20, 20	50, 20	100, 20
10, 30	20, 30	50, 30	100, 30

Model will be trained with all these combinations and it selects the best ones.

Randomised Search CV

[illegible] the one best combination

It select
among randomly selected
combinations

→ 10

Among 10 random combination
it will select the best one.

(a) Grid takes time but it is best
(b) Randomised faster but we
miss important ones.